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MONGO -DB CRUD OPERATION:

```
import com.mongodb.client.FindIterable;
import com.mongodb.client.MongoClient;
import com.mongodb.client.MongoClients;
import com.mongodb.client.MongoCollection;
import com.mongodb.client.MongoDatabase;
import com.mongodb.client.model.Filters;
import com.mongodb.client.model.Updates;
import org.bson.Document; public class Test2
{   public static void main(String[] args) {
        // Connect to MongoDB
        MongoClient mongoClient =
MongoClients.create("mongodb://localhost:27017");
        // Access Database and Collection
        MongoDatabase database = mongoClient.getDatabase("CollegeDB");
        MongoCollection<Document> collection = database.getCollection("Students");
        // =====
        // CREATE (Insert 5 Students)
        // =====      collection.insertOne(new
Document("id", 101)
        .append("name", "Aarav")
        .append("department", "CSE")
.append("cgpa", 8.5));      collection.insertOne(new
Document("id", 102)
        .append("name", "Priya")
        .append("department", "IT")
.append("cgpa", 9.1));      collection.insertOne(new
Document("id", 103)
        .append("name", "Rahul")
        .append("department", "ECE")
```

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```
.append("cgpa", 8.0));    collection.insertOne(new
Document("id", 104)
    .append("name", "Sneha")
    .append("department", "ME")
    .append("cgpa", 8.8));

collection.insertOne(new Document("id", 105)
    .append("name", "Vikram")
    .append("department", "CIVIL")
    .append("cgpa", 7.9));

System.out.println("5 Documents Inserted Successfully");
// =====
// READ (Display All Students)
// =====
System.out.println("\nStudent Records:");

FindIterable<Document> students = collection.find();    for
(Document doc : students) {
    System.out.println(doc.toJson());
}
// =====
// UPDATE
// Update CGPA of Student ID 103
// =====
collection.updateOne(
    Filters.eq("id", 103),
    Updates.set("cgpa", 9.3)
);

System.out.println("\nStudent ID 103 Updated Successfully");    //
```

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```
// READ Updated Record

// ===== Document
updatedStudent = collection.find(Filters.eq("id",
103)).first();
System.out.println("\nUpdated Record:");    if
(updatedStudent != null) {
    System.out.println(updatedStudent.toJson());
}
// =====
// DELETE
// Delete Student ID 105
// =====
collection.deleteOne(Filters.eq("id", 105));
System.out.println("\nStudent ID 105 Deleted Successfully");
// =====
// DISPLAY FINAL RECORDS
// =====
System.out.println("\nFinal Student Records:");    for
(Document doc : collection.find()) {
    System.out.println(doc.toJson());
}
// Close Connection    MongoClient.close();
}
```

```
For Many: package mongodb2;
import com.mongodb.client.FindIterable;
import
com.mongodb.client.MongoClient;
import
com.mongodb.client.MongoClients;
```

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```
import
com.mongodb.client.MongoCollection;
import
com.mongodb.client.MongoDatabase;
import
com.mongodb.client.model.Filters;
import
com.mongodb.client.model.Updates;
import    java.util.Arrays;    import
org.bson.Document; public class Test2 {
public static void main(String[] args) {
    // Connect to MongoDB
    MongoClient mongoClient =
MongoClients.create("mongodb://localhost:27017");
    // Access Database and Collection
    MongoDBDatabase database = mongoClient.getDatabase("CollegeDB");
    MongoCollection<Document> collection = database.getCollection("Students")
    // CREATE (Insert 5 Students)
collection.insertMany(Arrays.asList(        new
Document("id", 101)
        .append("name", "Aarav")
        .append("department", "CSE")
        .append("cgpa", 8.5),
new Document("id", 102)
        .append("name", "Priya")
        .append("department", "IT")
        .append("cgpa", 9.1),        new
Document("id", 103)
        .append("name", "Rahul")
        .append("department", "CSE")
```

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```
.append("cgpa", 8.0),          new
Document("id", 104)

    .append("name", "Sneha")
    .append("department", "ME")
.append("cgpa", 8.8),          new
Document("id", 105)

    .append("name", "Vikram")
    .append("department", "CSE")
    .append("cgpa", 7.9));

System.out.println("5 Documents Inserted Successfully");

// READ (Display All Students)
System.out.println("\nStudent Records:");    for
(Document doc : collection.find()) {
    System.out.println(doc.toJson());
}

// UPDATE    collection.updateMany(
    Filters.eq("department", "CSE"),
    Updates.set("department", "Computer Science")
);

System.out.println("\nMultiple Documents Updated Successfully");    //
READ Updated Record

System.out.println("\nRecords After Update:");    for
(Document doc : collection.find()) {
    System.out.println(doc.toJson());
}

// Delete Many    collection.deleteMany(
    Filters.eq("department", "Computer Science")
);

System.out.println("\nMultiple Documents Deleted Successfully");

// DISPLAY FINAL RECORDS
```

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```
        System.out.println("\nFinal Student Records:");    for
(Document doc : collection.find()) {
    System.out.println(doc.toJson());
}
// Close Connection    mongoClient.close();
}
}
```